```r
#file A3_aXXXXXXX.R
#Example R script for Assignment 3 (practica1)


## Read csv file
drill_dat <- read.csv("Drill.csv",header=T)


## Part a)
# ii
table(drill_dat$Region)

# iii
mean(drill_dat$Time_hr[drill_dat$Region%in%"Antarctic"]) #Antarctic
sd(drill_dat$Time_hr[drill_dat$Region%in%"Antarctic"])
mean(drill_dat$Time_hr[drill_dat$Region%in%"Pacific"])   #Pacific
sd(drill_dat$Time_hr[drill_dat$Region%in%"Pacific"])

# iv
var.test(Time_hr ~ Region, data=drill_dat) #F-test for equal variance

# v
t.test(Time_hr ~ Region,alternative="greater", data=drill_dat)


## Part b)
# i)
welch_t <- function(mu1,mu2,sd1,sd2,n1,n2){
    t_stat <- (mu1-mu2)/sqrt(((sd1^2)/n1)+((sd2^2)/n2))
    v1 <- n1-1
    v2 <- n2-1
    welch_df <- ((((sd1^2)/n1)+((sd2^2)/n2))^2)/((((sd1^4)/((n1^2)*v1))+(sd2^4)/((n2^2)*v2)))
    p_val <- pt(t_stat,welch_df,lower.tail=F)
    return(data.frame("t_statistic"=t_stat,"DF"=welch_df,"P"=p_val))
}


# ii)
welch_t(27.38,25,4.48,2.14,60,38)
welch_t(mean(drill_dat$Time_hr[drill_dat$Region%in%"Antarctic"]),25,sd(drill_dat$Time_hr[drill_dat$Region%in%"Antarctic"]),2.14,60,38)

# iv)
crit_region <- qt(1-0.05,90.42)
welch_df <- 90.42
t_stat <- 3.52

library(ggplot2)
ggplot(data.frame(c(0,0)), aes(c(-4,4))) +
    geom_area(stat="function",fun=dt,args=list(df=welch_df),fill="#00998a", xlim = c(crit_region,4)) + # shaded region
    stat_function(fun=dt,args=list(df=welch_df),size=1.2) + # density curve
    geom_vline(xintercept=t_stat,color="red",size=1.2,linetype="dashed") +
    labs(x="T",y="",title=paste0("t-distribution (df = ",welch_df,", alpha = ",0.05,")")) # axis labels
```